SIGNAL | Autonomous Financial Research ▶ WATCH THE 2-MIN DEMO 01 / 02

Computed, not retrieved.

SIGNAL finds where a company's numbers **persistently break three expectations** — its own history, its peers, and what its management said — and reports a finding only after it **survives a pre-registered attempt to disprove it**. *Find the breach. Trace the evidence. Try to break the hypothesis.*

01 THE PROBLEM

A non-obvious finding is, almost by definition, not written down anywhere. Retrieval can only return what someone has already said.

DASHBOARDS	show headline ratios — not the drivers underneath them.
SUMMARISERS	repeat the narrative; a "critic" prompt argues, it does not test.
SIGNAL	computes where **history ≠ peers ≠ narrative**, then tries to explain it away.

02 THE MECHANISM

three expectations

SELF	What its own history predicts	
PEER	What comparable firms show, same quarters	
NARRATIVE	What management said drives results	

EXPECTED history, peers, claims **VS** OBSERVED reconciled primary filings **=** BREACH a candidate, not yet a discovery

03 FROM CANDIDATE TO DISCOVERY

every gate must pass

G1	G2	G3	G4	G5	G6
Reconciled	**Corroborated**	**Specific**	**Survives tests**	**Predicts**	**Persistent**
Numbers satisfy accounting identities	≥2 independent source families	Not explained by peers	9 alternative explanations	A consequence it wasn't built from	Holds across time windows



Prototype · the three expectations behind one discovery

Illustrative prototype data · synthetic panel

04 WHY IT IS DIFFERENT

No LLM-typed numbers

Every figure is a computed value traced to a page and passage. The model only reads text and writes prose.

Independent evidence

Company, peers, regulator, rating agency. Three news copies of one press release count once.

Tested, not argued

Each counter-explanation is code over data, registered before the results are seen.

SIGNAL | Company in → auditable discovery out

▶ WATCH THE 2-MIN DEMO

05 RESEARCH WORKFLOW

target 5–7 minutes live

01	02	03	04	05	06	07	08
INGEST	**NORMALIZE**	**EXPECT**	**BREACH**	**TRIANGULATE**	**FALSIFY**	**RANK**	**DISCOVER**
Filings, transcripts, RBI, ratings	Typed facts, reconciled	History, peers, claims	Persistent surprise	Independent families	9 tests + corollaries	Gates + strength rule	8 required elements

Code decides — checks, statistics, tests, every number. **The LLM reads and writes** — claims, mechanisms, digit-free prose.

06 DISCOVERY FUNNEL

nothing hand-picked

143	Candidate breaches	all drivers
31	Structurally valid	reconciled, persistent
11	Corroborated	≥2 families
4	Company-specific	vs peers
2	Survived falsification	9 tests
1	**Discovery**	strength: strong

Every rejection is shown with its reason. ILLUSTRATIVE COUNTS

07 TRY TO BREAK THE DISCOVERY

pre-registered

AE1	**Sector-wide effect**	company vs same-quarter peers
AE2	**Regulatory / macro**	difference around dated events
AE3	**Accounting artefact**	policy, merger, restatement
AE4	**Denominator effect**	lagged ratios
AE5	**One-off / window**	specification curve
AE6	**Already disclosed**	claims + press coverage
AE7	**Data error**	second-document check
AE8	**Other mechanism**	predictions on unused data
AE9	**Counter-evidence**	bounded independent search

The strongest surviving alternative becomes the **counterargument**; strength is a **deterministic rule**, not an LLM opinion.

DISCOVERY 01 OF 1 · ASSET QUALITY · BANK ARCHETYPE

Over six quarters the subject bank's gross NPA ratio fell from 2.31% to 1.62% — but 71% of all NPA reductions were write-offs, while its fresh-slippage ratio rose from 1.40% to 2.20% as the peer median stayed flat at 1.44%. Management credited collections in 5 of 6 earnings calls.

The headline asset-quality improvement is real but is not what management says it is: it is write-off-led, not collections-led, and it coincides with a company-specific rise in fresh slippages that peers do not share.

✓ DISCOVERY · ✓ WHY IT MATTERS · ✓ EVIDENCE · ✓ TIMELINE · ✓ NUMBERS · ✓ COUNTERARGUMENT · ✓ FALSIFICATION CONDITION · ✓ EVIDENCE STRENGTH

EVIDENCE STRENGTH

STRONG ▬▬▬

Falsification tests survived	8 / 9
Independent source families	3
Persistence	5 quarters
Peer-specific	outside IQR 4 of 4Q
Visibility gap	4 quarters
Candidates considered	143

Prototype · every underlined claim opens its evidence trail

Illustrative output · not a finding about any company

08 EVERY NUMBER, TRACEABLE

VALUE	**+₹14,953 cr** fresh slippages, six quarters
FORMULA	open + slippages - upgrades - recoveries - write-offs = close ✓
SOURCE	Quarterly results · notes · p. 37
PASSAGE	Additions — fresh slippages 3,266

09 GENERALIZATION

archetype, not company

BANKS NPA flows, deposits, capital	**NBFCs** AUM, funding mix, Stage 2
INSURERS APE mix, margins	**INFRA · HOLDCOS** generic identities

Benchmarked on **held-out NIFTY** and **non-NIFTY** companies, plus **as-of backtests**.

10 BUILD PLAN

now: prototype

08–15 Oct	Ingestion, reconciliation, expectation panels
14–19 Oct	Falsification, provenance, live dashboard
19–25 Oct	Benchmark, hardening, live rehearsals